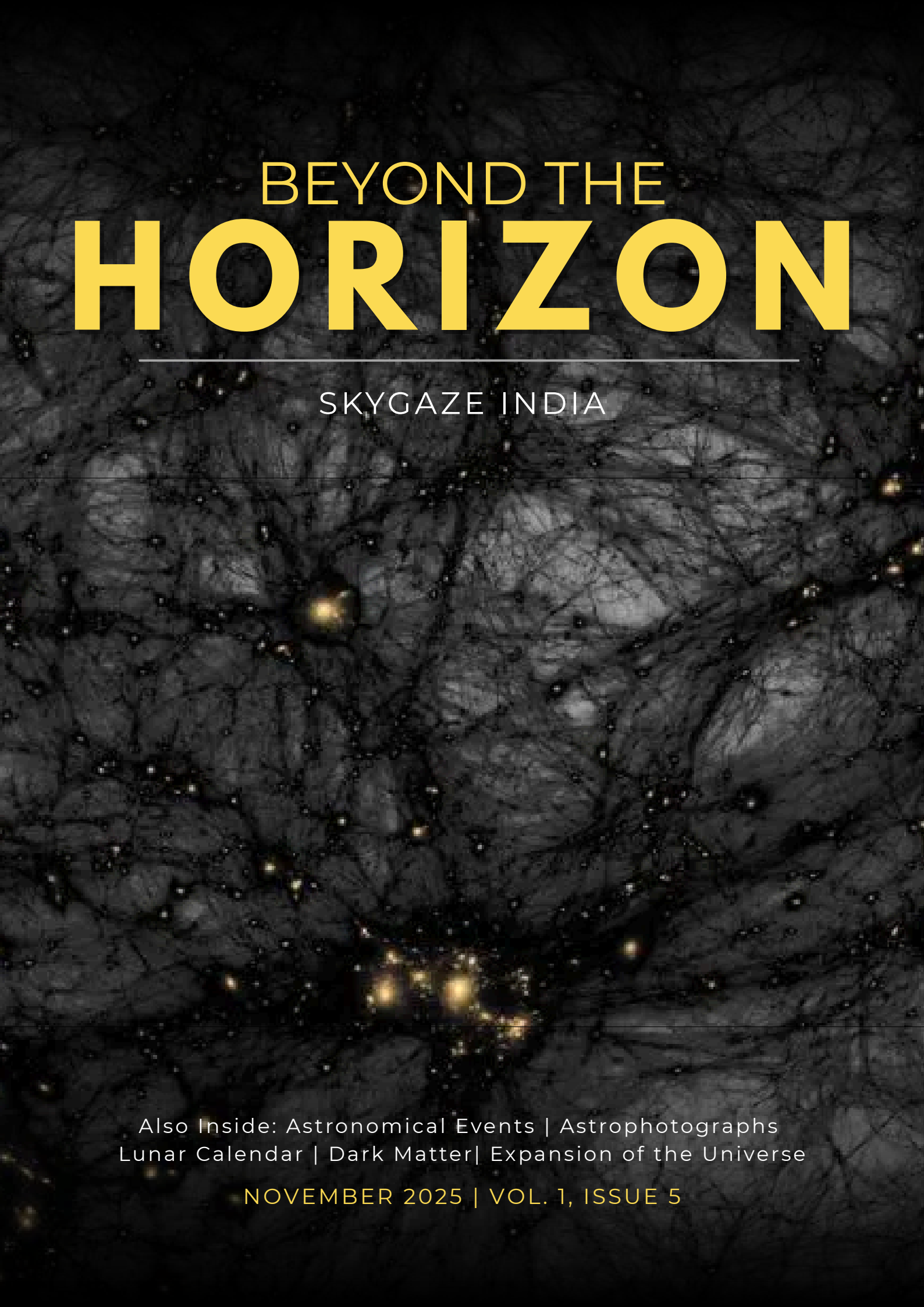




BEYOND THE HORIZON

SKYGAZE INDIA

Also Inside: Astronomical Events | Astrophotographs
Lunar Calendar | Dark Matter | Expansion of the Universe

NOVEMBER 2025 | VOL. 1, ISSUE 5

INTRODUCTION

Beyond the Horizon, Skygaze India's monthly publication, is meticulously curated to stimulate intellectual curiosity and foster deeper public engagement with astronomy and space science.

Each issue presents an amalgamation of celestial event highlights, exemplary astrophotography features, perspicacious articles, and exclusive content contributed by our community of astronomy enthusiasts. Our objective is to make space science both accessible and profoundly inspiring.

OUR TEAM

Founder & Editor in Chief

Harsh

Contributors

Hams Gadalla, Egypt

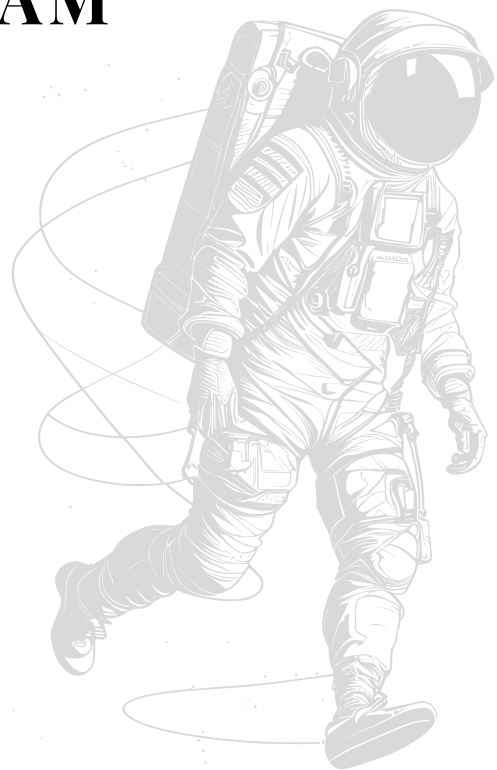
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"At Skygaze India, our mission is to make space science accessible—to bring the cosmos closer by turning complex knowledge into everyday curiosity for all."

Harsh
Founder & CEO

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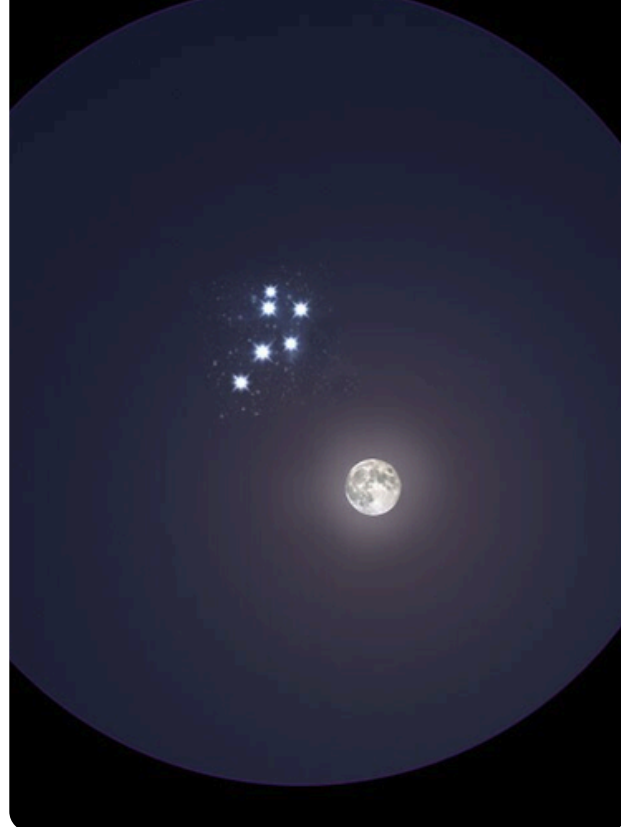
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ASTRONOMICAL EVENTS



NOVEMBER 5: BEAVER FULL MOON

The Beaver Moon will illuminate the skies this month. Traditionally, it marked the time when beavers built their winter dams. Expect a bright, crisp full moon—an ideal moment for moonlit photography or simply enjoying the serene glow of autumn nights.

NOVEMBER 6: CLOSE CONJUNCTION OF PLEIADES AND MOON

A captivating celestial pairing occurs as the Moon passes close to the Pleiades star cluster in Taurus. Visible for most of the night, this conjunction offers a spectacular view to the naked eye and an even more enchanting scene through binoculars or a telescope.

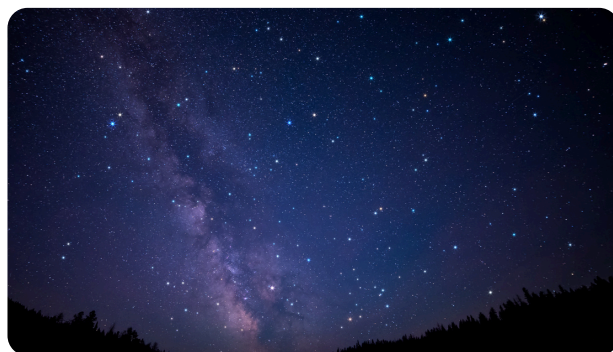


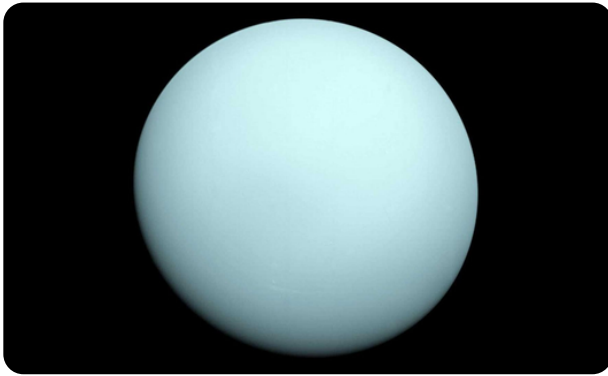
NOVEMBER 17-18: LEONIDS METEOR SHOWER

The Leonids, one of the most famous meteor showers, will peak on these nights. With up to 15 meteors per hour under dark skies, observers can look forward to bright, fast streaks radiating from the constellation Leo. The nearly new Moon ensures dark skies and ideal viewing conditions this year.

NOVEMBER 20: NEW MOON

The sky will turn its darkest as the Moon reaches its new phase. This is the best opportunity of the month to observe faint deep-sky objects—such as galaxies and nebulae—without interference from moonlight. The absence of lunar glow makes it ideal for stargazing and astrophotography under truly pristine skies.

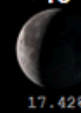
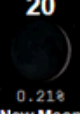
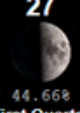




NOVEMBER 21: URANUS OPPOSITION

Uranus will be at its closest approach to Earth and fully illuminated by the Sun. Appearing as a small bluish-green disk, it can be spotted with binoculars or a small telescope in the constellation Aries. A rare treat for planetary observers! Don't miss the chance to witness this distant ice giant at its brightest.

LUNAR CALENDAR

Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
						01  80.78% Waxing Gibbous 10.5 day
02  88.98% Waxing Gibbous 11.58 day	03  95.28% Waxing Gibbous 12.71 day	04  99.07% Full Moon 13.86 day	05  99.92% Full Moon 15.03 day	06  97.7% Full Moon 16.2 day	07  92.62% Waning Gibbous 17.35 day	08  85.15% Waning Gibbous 18.48 day
09  75.97% Waning Gibbous 19.58 day	10  65.74% Last Quarter 20.64 day	11  55.1% Last Quarter 21.67 day	12  44.57% Last Quarter 22.66 day	13  34.57% Last Quarter 23.62 day	14  25.43% Waning Crescent 24.56 day	15  17.42% Waning Crescent 25.48 day
16  10.74% Waning Crescent 26.39 day	17  5.56% Waning Crescent 27.29 day	18  2.03% New Moon 28.19 day	19  0.23% New Moon 29.08 day	20  0.21% New Moon 0.44 day	21  1.98% New Moon 1.33 day	22  5.49% Waxing Crescent 2.22 day
23  10.66% Waxing Crescent 3.13 day	24  17.35% Waxing Crescent 4.04 day	25  25.4% Waxing Crescent 4.97 day	26  34.59% First Quarter 5.91 day	27  44.66% First Quarter 6.88 day	28  55.25% First Quarter 7.88 day	29  65.93% First Quarter 8.91 day
30  76.19% Waxing Gibbous 9.97 day						

ASTROPHOTOGRAPHS



Supermoon captured by Ms. Hams Gadalla, Founder at Astrotopia Club Junior, Egypt



The Moon captured by Ms. Jheel Jain, Aspiring Astrophysicist, Rajasthan, India

DARK MATTER VS. DARK ENERGY:

THE COSMIC BALANCE

Aspect	Dark Matter	Dark Energy
What It Is	Invisible mass that creates gravity	A mysterious force causing the universe to expand faster
Role in the Universe	Holds galaxies together and shapes cosmic structure	Drives accelerated expansion of space
Interaction with Light	Does not emit, absorb, or reflect light	Does not interact with light or matter (except through expansion)
Interaction with Matter	Affects matter through gravity	Does not clump or form structures — acts uniformly
Evidence for Existence	Galaxy rotation curves, gravitational lensing, cosmic web simulations	Dimming of distant supernovae, CMB measurements, cosmic expansion data
Percentage of the Universe	~27% of the total energy-density	~68% of the total energy-density
How It Behaves	Clumps around galaxies in “halos”	Stretches space evenly everywhere
What It Might Be Made Of	WIMPs, axions, sterile neutrinos	Possibly a property of space itself (cosmological constant) or “quintessence”
Effect on the Future of the Universe	Influences galaxy formation and stability	Determines long-term expansion: slow, steady, or runaway expansion
Why It’s a Mystery	We know its gravitational pull, but not its particle nature	We observe its effects but have no direct theory explaining its origin
Discovered / Proposed	Idea strengthened in the 1970s through galaxy rotation studies	Recognized in 1998 through supernova observations
Can It Be Detected Directly?	Scientists are actively searching via detectors and space signals	Cannot be detected directly — inferred from cosmic expansion

WHAT IS DARK MATTER REALLY MADE OF? — EXPLORING LEADING THEORIES

Dark matter is one of the biggest cosmic mysteries of our time. Even though we cannot see it, scientists are almost certain that it exists because the universe behaves as if a huge amount of invisible mass is present. Everything we see — stars, planets, galaxies, and all visible matter — makes up less than five percent of the universe. The rest is hidden, and a major part of that hidden universe is dark matter, a substance that neither glows nor absorbs light but has tremendous gravitational influence.

We know dark matter exists because of three key observations made over the past century:

- **Galaxies rotate too fast:**

Stars at the edges of galaxies move so quickly that they should fly off into space, but they don't. Something unseen is providing the extra gravity needed to hold them in place.

- **Light bends more than expected:**

When light from distant galaxies passes near large clusters, it bends more strongly than what visible matter alone can explain. This effect, called gravitational lensing, indicates the presence of hidden mass.

- **The universe's large-scale structure needs it:**

Computer simulations of the universe only match reality when dark matter is included. It acts like an invisible framework on which galaxies are built.

With strong evidence pointing to its existence, the natural question is: **What is dark matter made of?** Scientists have proposed several candidates, but three leading theories stand out.



Dark matter filaments connecting galaxies across the cosmic web.

1. WIMPs — Weakly Interacting Massive Particles

WIMPs have been the most popular dark matter candidates for decades. They are theoretical particles that have mass but interact so weakly with normal matter that they pass through it unnoticed.

WIMPs are compelling because:

- They would have formed naturally in the early universe.
- Their predicted abundance matches the observed amount of dark matter.
- They fit well into theories of physics beyond the Standard Model.

Experiments deep underground are attempting to detect the rare collisions between WIMPs and normal atoms. So far, none have been confirmed, but the search continues globally.

2. Axions — Ultra-Light, Wave-Like Particles

Axions are another strong possibility. Unlike WIMPs, axions are extremely light — a billion times lighter than electrons. They are predicted to behave more like waves spreading across the universe rather than individual particles.

Axions are exciting because:

- They could form vast, invisible “axion clouds” around galaxies.
- They interact so weakly that they could easily remain hidden.
- They naturally solve certain physics problems unrelated to dark matter.

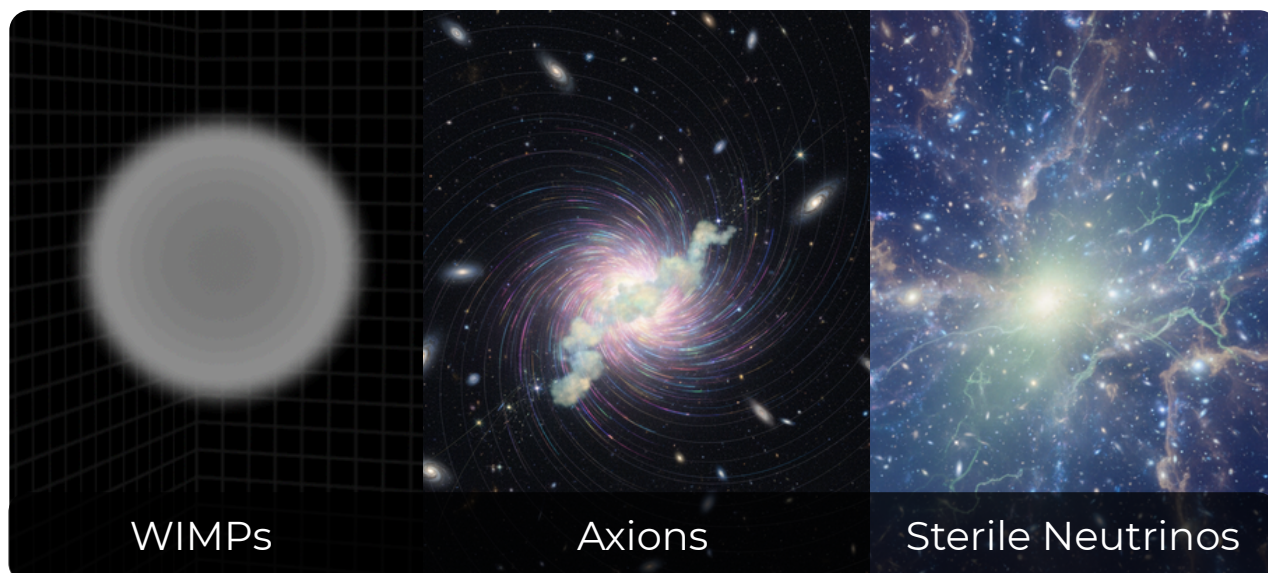
Scientists are designing experiments where axions might convert into faint photons inside strong magnetic fields. While hints have appeared, no direct evidence has yet been found.

3. Sterile Neutrinos — The Hidden Cousins

We already know about three kinds of neutrinos, which are tiny, fast-moving, and extremely difficult to detect. A sterile neutrino would be an even more elusive version — one that does not interact through any force except gravity. Sterile neutrinos are compelling because:

- They might explain strange patterns in how galaxies form.
- They may be connected to mysterious X-ray signals seen in space.
- They fit naturally into several advanced particle physics theories.

However, studies have shown mixed results, so scientists continue to investigate.



Leading dark matter candidates

So, What Is Dark Matter Really Made Of?

The truth is that no one knows — yet. But that is what makes dark matter one of the most thrilling fields in modern science. We know it outweighs normal matter, shapes galaxies, and influences the evolution of the universe. What we don't know is what type of particle — WIMPs, axions, sterile neutrinos, or something entirely new — is responsible for this invisible cosmic glue.

Researchers worldwide are building sensitive detectors, analyzing deep-space data, and running massive simulations to uncover the answer. When the true nature of dark matter is finally discovered, it may open the door to new physics and revolutionize our understanding of the universe.

Dark matter reminds us that the cosmos is full of mystery. Most of the universe is still unseen, waiting to be explored — and that sense of wonder is what drives astronomy forward.

-Harsh
Founder, Skygaze India

CONTRIBUTE TO BEYOND THE HORIZON!

Have an astronomy article, story, or thought to share?
We're inviting written submissions for our next publication!

Submit via email on support@skygazeindia.com



EXPANSION OF THE UNIVERSE:

WHY SPACE ITSELF IS STRETCHING

For a long time, humanity believed the universe was static eternal and unchanging. Yet, a few curious minds dared to question this still image of existence: Could a universe so vast truly be motionless? Or was there something deeper, hidden within the fabric of space itself, that we had yet to understand? As the 20th century unfolded, everything changed. Thanks to remarkable advances in astronomy and the study of light from distant **galaxies**, scientists discovered that space is not static but expanding and it has been doing so since its very beginning, around 13.8 billion years ago, according to the **Big Bang theory**.

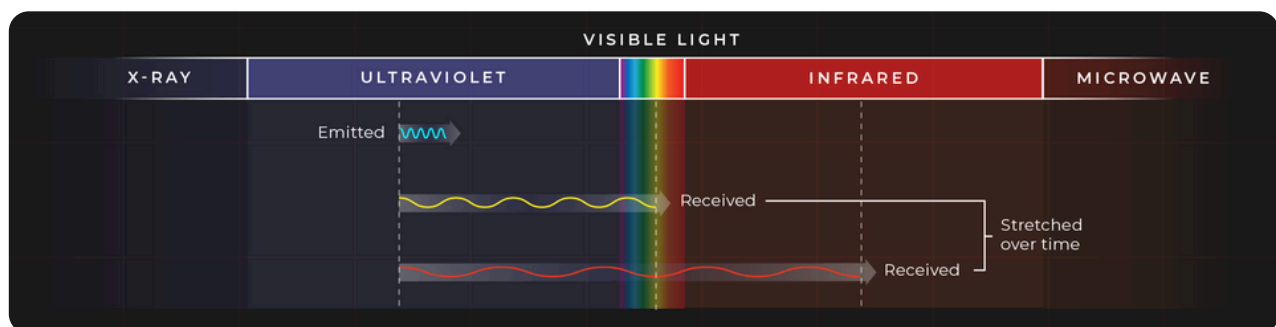
This revelation transformed our understanding of the cosmos. Galaxies and stars are not fixed in place; they are moving away from each other in a coordinated, measurable pattern. This was not speculation it was a conclusion drawn from precise observational evidence that reshaped modern cosmology.

And so, a fascinating question emerged: How did scientists realize that the universe is expanding? What kind of evidence turned this idea into one of the cornerstones of modern astrophysics?

1. The Redshift of Light

One of the most compelling pieces of evidence comes from what scientists call the "redshift" the stretching of light waves as galaxies move away from us. When astronomers analyzed light from distant galaxies, they noticed that its wavelengths shifted toward the red end of the electromagnetic spectrum — toward longer, lower-energy waves. This phenomenon, known as the "Doppler Effect", shows that waves stretch when their source moves away from the observer. In 1929, American astronomer "Edwin Hubble" made a breakthrough: he found that the farther away a galaxy is, the faster it is receding from Earth. This relationship, now known as "Hubble's Law", provided the first direct evidence that space itself is expanding, not just that galaxies are moving through empty space.

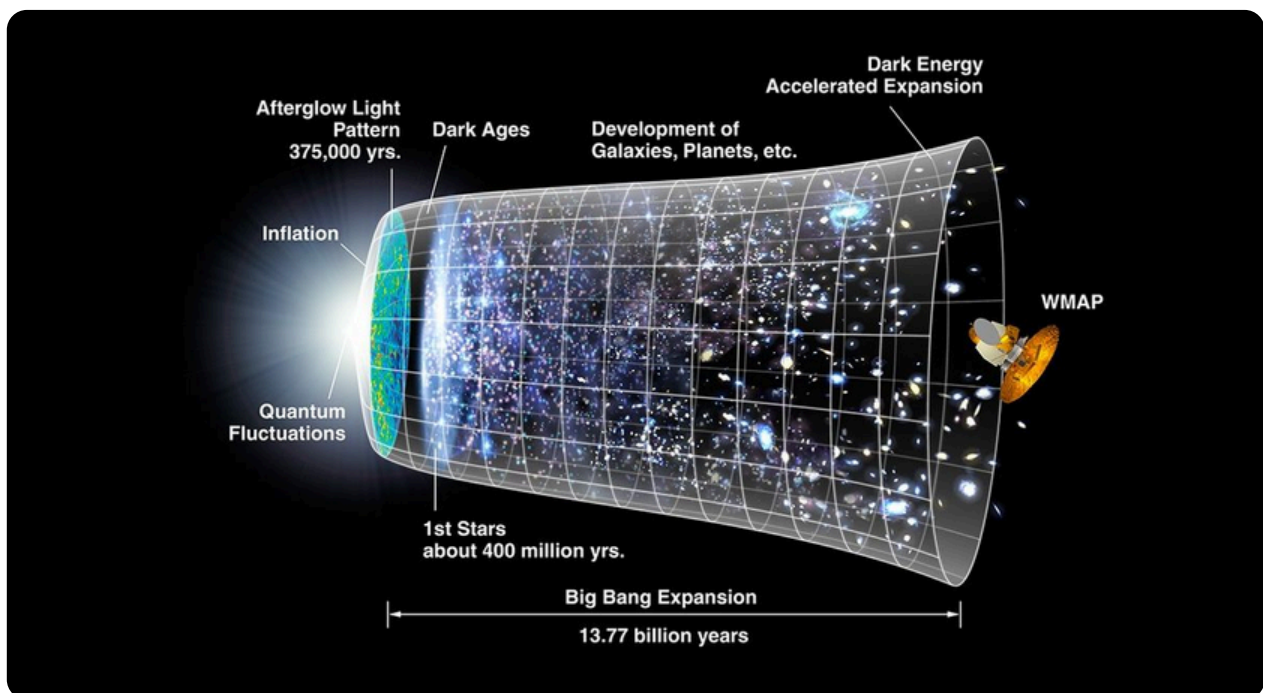
That single discovery revolutionized our understanding of the universe's structure and became a foundational pillar for modern cosmological models, especially the "Big Bang theory", which describes how the universe originated and evolved through time.



Redshift in Light, NASA

2. The Cosmic Microwave Background: The Oldest Light in the Universe

Another line of evidence comes from a faint, ancient glow the "**Cosmic Microwave Background (CMB)**" often described as the afterglow of creation. This radiation serves as strong proof that the universe once existed in a state of extreme heat and density, just as the Big Bang theory predicts. Roughly 380,000 years after the Big Bang, the universe cooled enough for the first atoms of hydrogen and helium to form. At that moment, photons — particles of light — were finally able to travel freely through space without being absorbed or scattered. That same light has been journeying across the cosmos ever since, stretching along with space itself until its wavelength shifted into the microwave range we detect today.



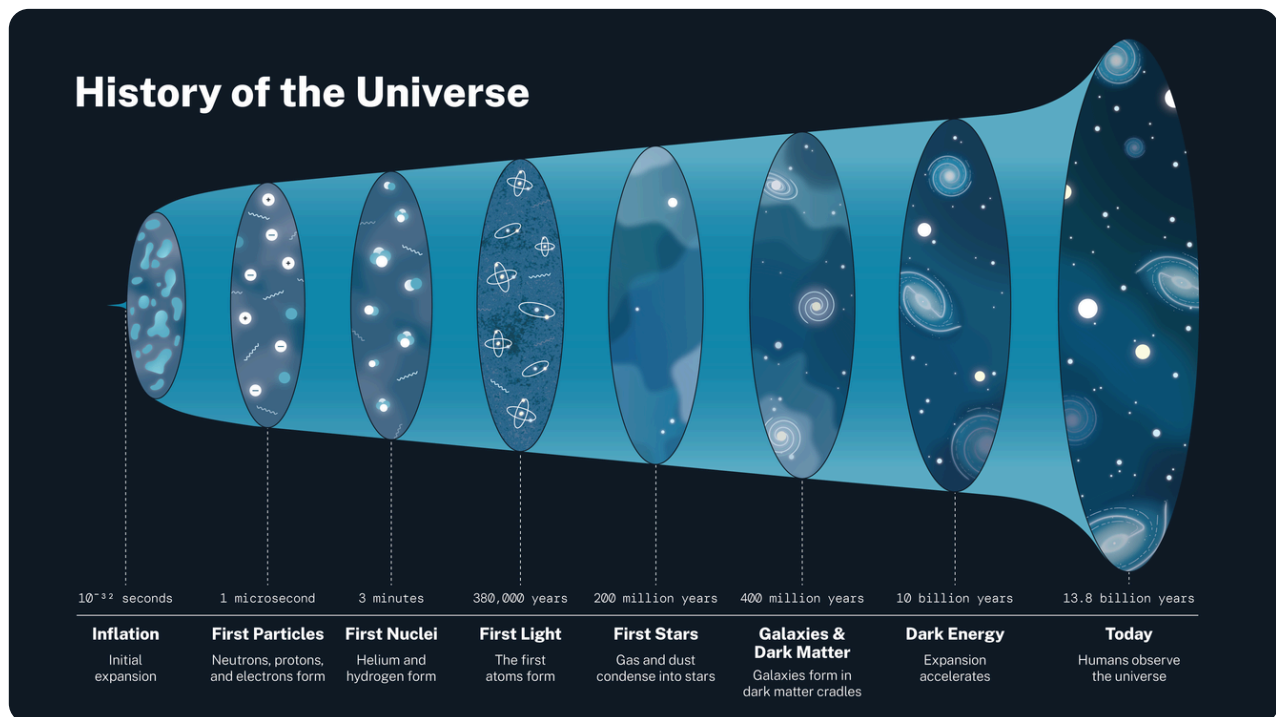
Timeline of the Universe — from the Big Bang to today, NASA

In 1965, two American radio astronomers, "Arno Penzias" and "Robert Wilson", accidentally discovered this faint, uniform background noise while working at Bell Laboratories. No matter where they pointed their antenna, the signal was always there — soft, steady, and mysterious. Later studies confirmed that this was no ordinary noise but the "thermal echo of the early universe" the leftover radiation from the Big Bang itself.

Today, the CMB measures about 2.7 Kelvin, astonishingly uniform with tiny fluctuations no greater than a few thousandths of a degree. These slight variations allowed scientists to study the universe's earliest structure and trace how galaxies and clusters later formed. The very existence and gradual stretching of this ancient light provide direct evidence that space has been expanding continuously since the dawn of time. The CMB stands as a silent witness to the universe's history.

3. The Ratios of Elements: A Nuclear Proof of Expansion

Finally, there is a nuclear fingerprint written into the matter around us. Calculations based on the Big Bang model predict that roughly 75% of the visible matter in the universe is hydrogen, while helium makes up about 25%, with trace amounts of lithium and heavier elements (Peebles, 1966; Wagoner et al, 1967). These predictions were later confirmed through observations of stars, nebulae, and precise measurements from satellites such as WMAP and Planck, which studied the Cosmic Microwave Background in remarkable detail (Planck Collaboration, 2018). The observed ratios match exactly what Big Bang nucleosynthesis calculations predict a perfect agreement between theory and observation. This alignment confirms that the universe began in a hot, dense state and has been expanding ever since.



Timeline of cosmic evolution highlighting the era of element formation, NASA

Conclusion

These discoveries, each building upon the other, paint a breathtaking portrait of our expanding universe. They remind us that every faint signal from the distant cosmos carries a story a whisper from the dawn of time. The expansion of the universe is not just a scientific fact; it's a reflection of cosmic evolution, of a universe that continues to grow and transform. And as we gaze into the vastness above, one question remains open:

Will this expansion go on forever, or does the universe hold a secret ending yet to be revealed? Whatever the answer, the search for it is what keeps humanity looking up still curious, still wondering.

-Assil Bedra

Student, Khadraoui Brothers High School

3I/ATLAS:

THE THIRD CONFIRMED INTERSTELLAR VISITOR

Discovery & Classification

Discovered on 1 July 2025 by the ATLAS sky-survey in Chile, 3I/ATLAS is the third known object from outside our Solar System to pass through — following 1I/'Oumuamua and 2I/Borisov. Its orbit is hyperbolic (eccentricity > 1), confirming its interstellar origin.

Its closest approach to the Sun (perihelion) occurred on 29 October 2025 at ≈ 1.36 AU, and it will come no closer than ~ 1.8 AU to Earth — meaning no threat to our planet.

Key Physical & Orbital Characteristics:

- Origin: presumably from the Milky Way's thin or thick disk, possibly over 7-10 billion years old.
- Nucleus: estimated diameter is uncertain but likely less than ~ 5 km, surrounded by a coma of dust and gas.
- Orbit: extremely high eccentricity, moving too fast to be bound by the Sun.



The interstellar comet 3I/ATLAS during its 2025 flyby

What Makes It Special:

- As an interstellar visitor, 3I/ATLAS offers a rare chance to study material formed outside our Solar System — a direct sample of alien space chemistry.
- Observations indicate unusual composition: high carbon-dioxide levels, low water ice relative to other comets, strong dust emission and nickel/cyanide gas signatures.
- Its size appears larger than previous interstellar objects, making it the largest known ISO so far.

Observational Highlights & Scientific Importance:

- Imaged by the Hubble Space Telescope on 21 July 2025, revealing a dusty tail and sun-facing plume.
- Observed by multiple facilities including the Very Large Telescope, James Webb Space Telescope, and Vera C. Rubin Observatory, enabling detailed investigation of its composition and behavior.

Because of its interstellar trajectory and composition differences, 3I/ATLAS is a key object for understanding how material from other star systems may differ from our own.

THIS MONTH IN INDIAN SPACE HISTORY

NOV, 2008 14	MOON IMPACT PROBE LUNAR DESCENT On 14 November 2008, the Moon Impact Probe separated from Chandrayaan-1 and made a controlled impact on the lunar surface. This milestone provided crucial data and marked one of India's earliest successful steps in lunar exploration.
NOV, 2013 05	MOON IMPACT PROBE (MIP) LUNAR DESCENT India successfully launched its first interplanetary mission, the Mars Orbiter Mission (Mangalyaan), on 5 November 2013. This historic mission made India the first Asian nation to reach Martian orbit and the first in the world to do so on its maiden attempt.
NOV, 2025 02	LAUNCH OF CMS-03 COMMUNICATIONS SATELLITE ISRO launched its heaviest communications satellite, CMS-03, on 2 November 2025 aboard the LVM3-M5 (GSLV-MkIII) rocket. This mission represented a significant advancement in India's satellite communication infrastructure and launch capability.

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We're bringing the universe closer to you!
Join us at Select City Mall for an unforgettable skygazing experience — explore the sky through powerful telescopes, learn fascinating cosmic facts, and see the stars like never before.

Discover. Observe. Wonder.

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